

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

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1. INTRODUCTION

1.1 Project Overview

The revised implementation treats support-ticket prioritization as a controlled decision pipeline rather than a single keyword lookup. A customer request enters Salesforce, the relevant Account and ticket context is collected, urgency and service-risk signals are evaluated, and the resulting priority is passed to an assignment and task-creation path. Agentforce provides the conversational entry point while Salesforce Flow remains responsible for deterministic execution.

1.2 Purpose

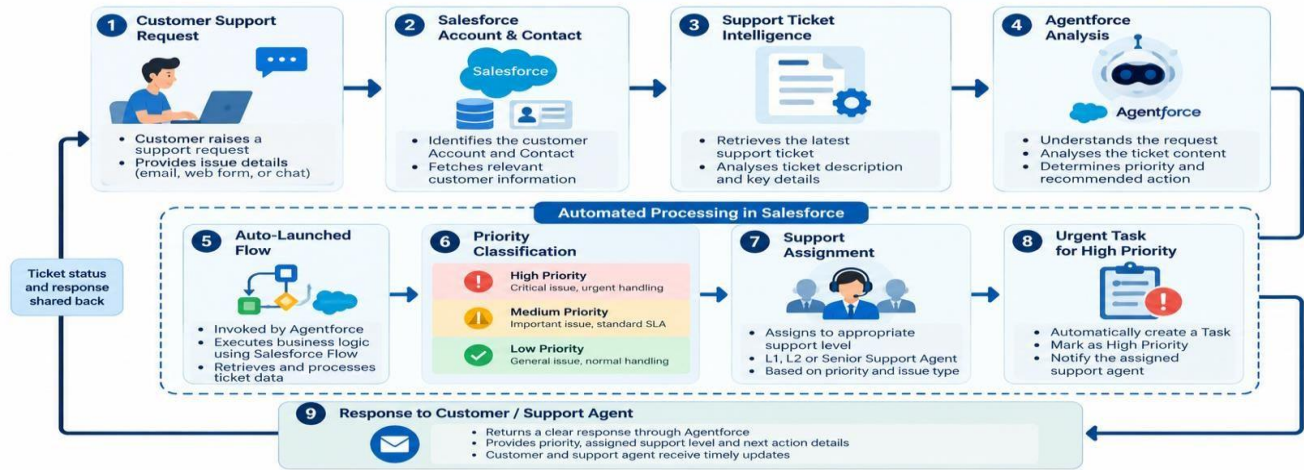
- Reduce the amount of repetitive triage performed by support staff.
- Provide a consistent priority decision from the same configured signals.
- Keep customer data, automation and action records inside the Salesforce operating context.
- Create an explicit follow-up task when the decision reaches the urgent band.
- Return a compact response that a support user can act on immediately.

1.3 Project Objectives

Objective	Updated Expected Result
Context enrichment	Resolve the customer record and retrieve the most relevant support ticket before evaluation.
Priority scoring	Convert urgency, impact and age signals into a High, Medium or Low decision band.
Automated action	Create a traceable follow-up task for the High band.
Assignment output	Return a configured support level or queue rather than an unstructured recommendation.
Conversational access	Allow Agentforce to request the same controlled workflow using a bounded action.

Customer Support Automation Ecosystem

(Ticket Priority Prediction and Automated Assignment using Salesforce and Agentforce)



2. IDEATION PHASE

2.1 Problem Statement

Support teams frequently receive requests with different levels of urgency, incomplete context and competing service deadlines. The design problem is not simply identifying a keyword; it is creating a repeatable path from customer context to a support action. The revised concept therefore separates interpretation from execution so that conversational AI can initiate a workflow without owning the final operational state.

Stakeholder	Need	Observed Friction	Revised System Response
Support Agent	See important work early	Manual scanning of descriptions	Priority band + action message
Customer	Receive appropriate attention	Urgent cases can join normal queues	Urgent path with task creation
Support Lead	Keep routing consistent	Different agents use different judgment	Configured assignment rules
Agentforce	Use trusted backend actions	Free-form actions can be unpredictable	Bounded Flow action with defined inputs

2.2 User Personas

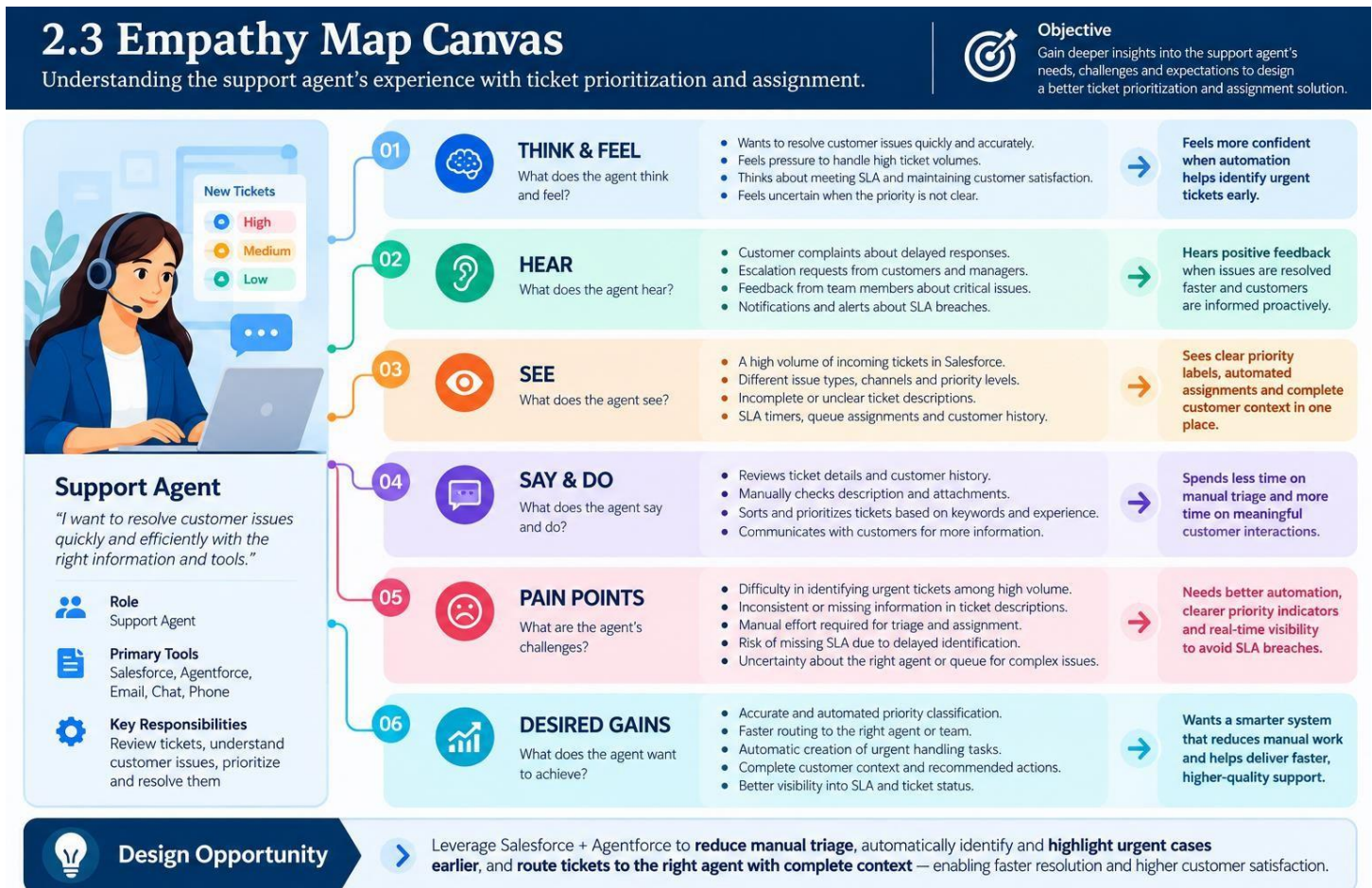
Persona	Primary Goal	System Interaction
Support Agent	Resolve the right work in the right order	Receives priority, assignment and next action
Customer	Get the issue recognized and tracked	Creates or updates a support request
Support Lead	Control queue quality and exceptions	Reviews escalations, tasks and SLA risk
Agentforce	Interpret intent and invoke a safe action	Collects the account identifier and calls the Flow

2.2 Problem Statement

Understanding the real challenges faced by stakeholders in the customer support process



2.3 Empathy Map Canvas



2.4 Brainstorming and Idea Selection

The design exploration compared approaches by operational control, reuse of existing CRM data, traceability and ease of adoption. A standalone portal was not selected for the core workflow because it would introduce another operational surface. A purely conversational solution was also avoided because ticket updates and task creation require deterministic backend behavior.

- Manual register — minimal setup, but every priority decision remains human-driven.
- Spreadsheet rules — easy to modify, but weak for record-level control and auditability.
- Standalone support portal — stronger interface, but duplicates CRM context.
- Flow-first automation — reliable execution, but requires a user-facing entry point.
- Salesforce + Agentforce — combines conversational access with governed automation.

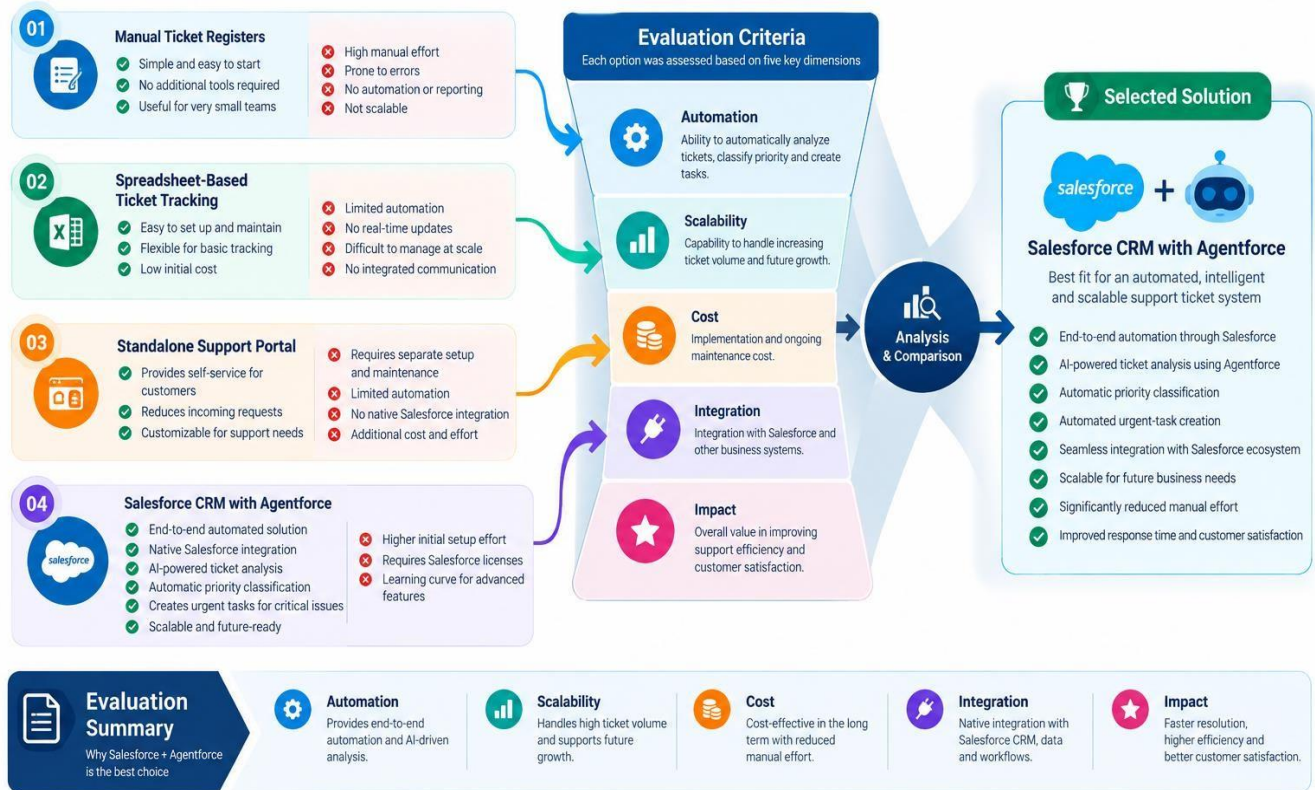
Brainstorming & Idea Prioritization

Evaluating solution options and selecting the best approach for an automated support ticket management system



Objective

Identify the most suitable solution for an automated ticket system based on key evaluation criteria and business needs.



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The revised journey begins with the customer request and ends with a visible operational action. Each transition has a clear responsibility: Salesforce stores context, the Flow performs deterministic evaluation, and Agentforce communicates the result.



The key checkpoints are context resolution, latest-ticket retrieval, signal evaluation, priority assignment, queue selection and action creation.

3.2 Functional Requirements

ID	Requirement	Description
FR-01	Ticket Record Management	Create and maintain the structured support-ticket record used by automation.
FR-02	Customer Association	Associate each ticket with an Account and, where available, a Contact.
FR-03	Relevant Ticket Retrieval	Retrieve the most relevant recent ticket for the requested customer context.
FR-04	Signal Evaluation	Evaluate urgency, operational impact and age-related indicators.
FR-05	Priority Decision	Return a High, Medium or Low priority band.
FR-06	Urgent Task Creation	Create a follow-up Task when the High band is selected.
FR-07	Assignment Output	Return the configured support level or queue.
FR-08	Agentforce Invocation	Allow a conversational request to trigger the workflow.
FR-09	Traceable Result	Return record identifiers and an action message for audit and follow-up.

3.3 Non-Functional Requirements

Quality	Requirement
Usability	Prompts and action outputs should be understandable to support users.
Reliability	Identical inputs and configured rules should produce consistent decisions.
Performance	Record lookup and Flow execution should complete within normal interaction time.
Security	Customer and ticket information must remain within authorized Salesforce access.
Maintainability	Priority bands and assignment rules should be editable without redesigning the interface.
Scalability	The workflow should support more tickets, queues and SLA conditions as the project grows.

3.4 Data Flow

Account context → Agentforce action → Flow input → customer lookup → ticket retrieval → signal evaluation → priority band → assignment/task action → Agentforce response.

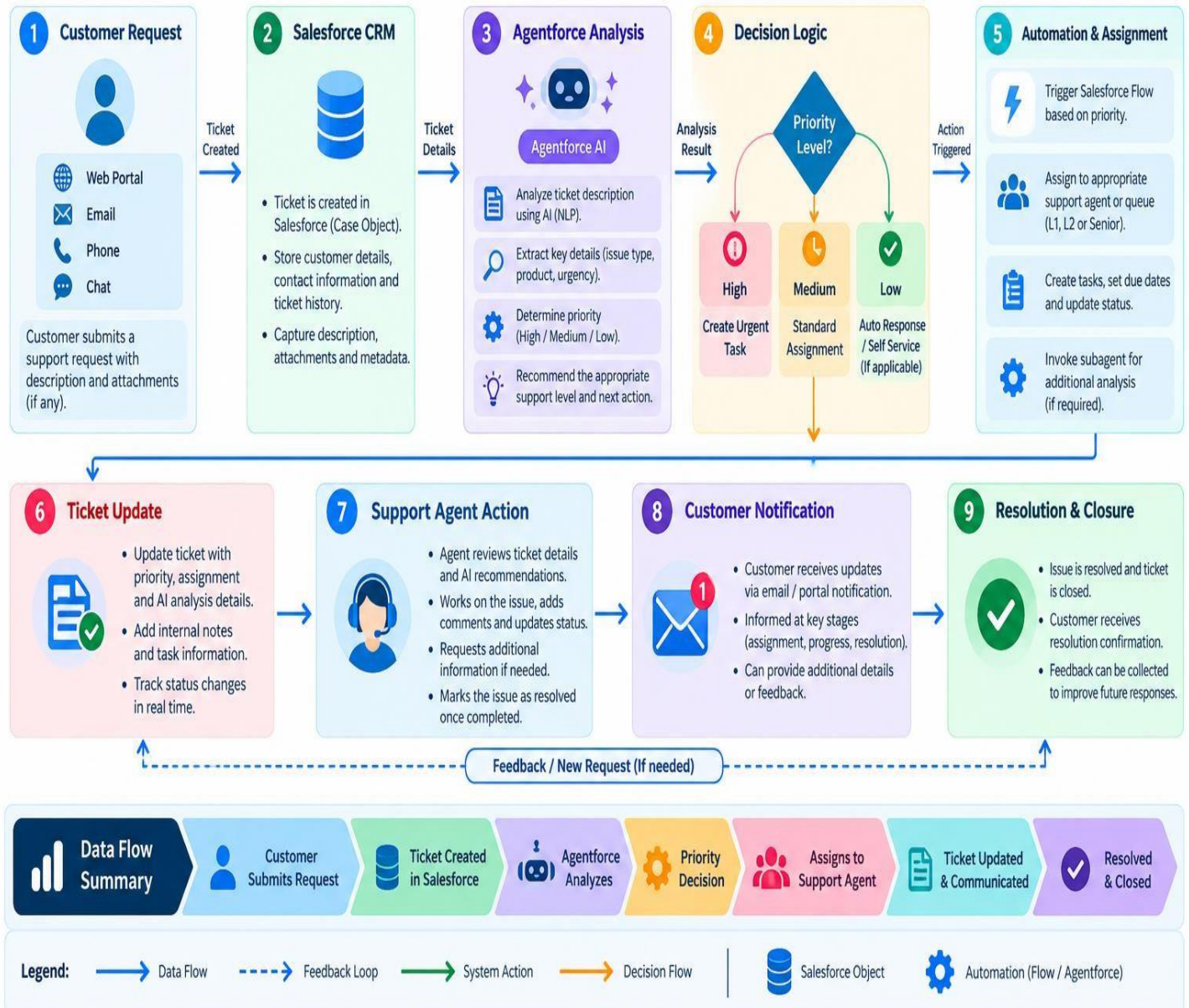
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Data Flow Diagram

Customer Support Ticket Automation with Agentforce – End to End Flow



Automate.
Prioritize.
Assign.
Resolve Faster.

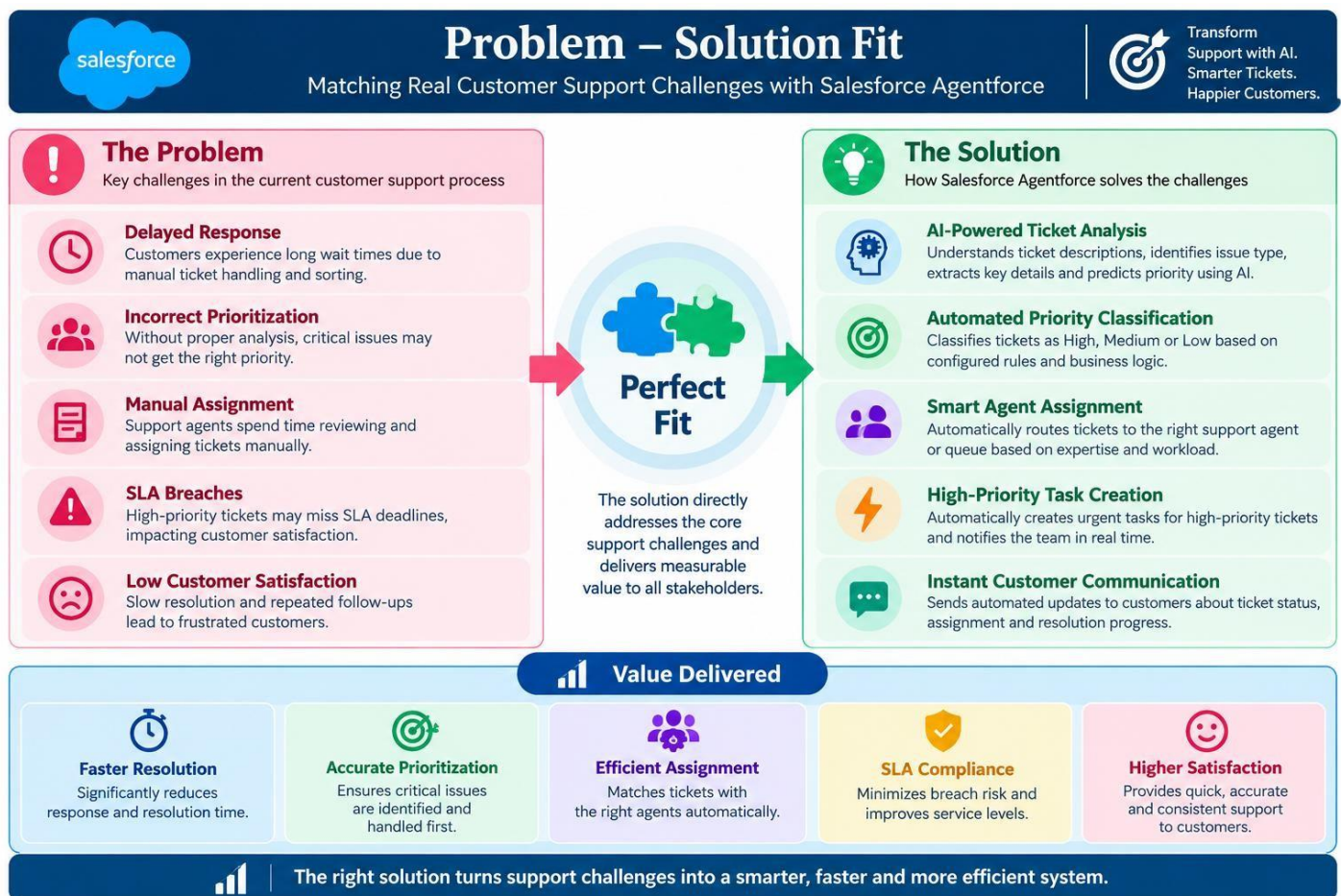


3.5 Technology Stack

Layer	Technology / Component
CRM Platform	Salesforce
Business Data	Account, Contact, Support Ticket Intelligence, User, Task
Automation	Auto-Launched Salesforce Flow
Conversational Layer	Agentforce
Decision Logic	Configured priority and assignment rules
Development	Salesforce DX and Salesforce CLI
Version Control	Git and GitHub

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit



Problem Area	Designed Solution	Resulting Behavior
Manual triage	Signal-based Flow evaluation	Priority is calculated automatically
Scattered customer context	Account + ticket retrieval in one path	Decision uses a consistent record context
Urgency can be missed	High-risk signal band	Urgent handling path becomes explicit
Manual follow-up	Task creation from the High path	Action becomes traceable
Inconsistent routing	Configured support-level output	Assignment follows the same rule set
Conversational access	Agentforce action	User can request analysis in natural language

4.2 Proposed Solution

The proposed solution combines a Salesforce custom object, an Auto-Launched Flow and an Agentforce subagent. Agentforce gathers the minimum required account input and invokes the action. The Flow resolves records, evaluates configured signals, assigns a priority band, creates urgent work when required and returns structured outputs. This separation keeps the conversation flexible while keeping business state deterministic.

4.3 Solution Architecture

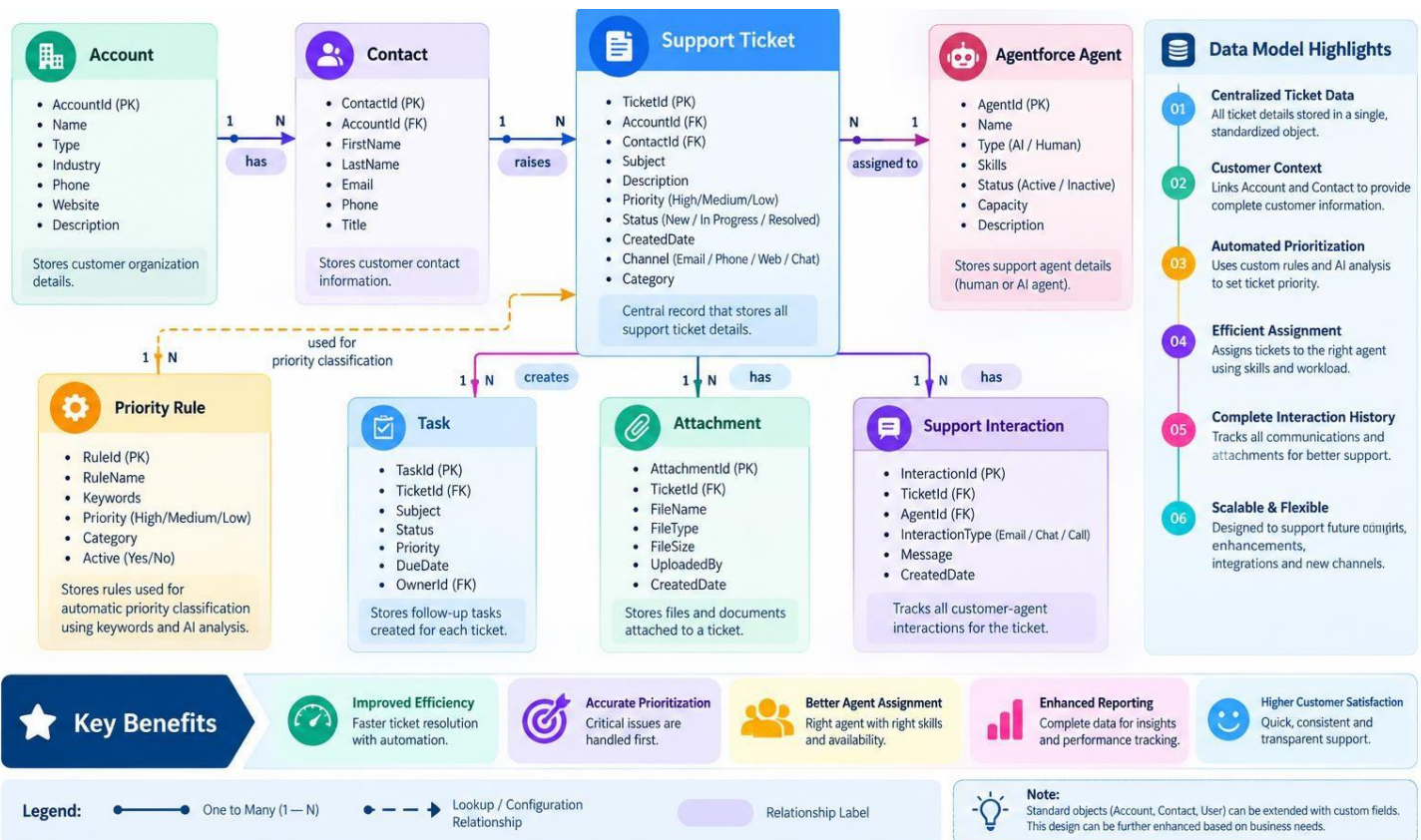


4.4 Data Model Design

The Support Ticket Intelligence record remains the central business object. Customer identity is represented through Account and Contact relationships, while operational fields capture issue context, priority, status, assignment and service risk. Task records provide the action trail for cases that need immediate attention. The model is intentionally modular so additional SLA or analytics fields can be introduced without changing the Agentforce interface.

4.5 Priority Decision Logic

The revised logic groups evidence into three decision bands instead of treating one phrase as the entire decision. A critical signal can move a case directly to High; a meaningful operational issue without critical evidence can result in Medium; and cases with no elevated signal remain Low. Ticket age and SLA risk can act as secondary escalation signals.



The decision should remain explainable: the output records the selected band and the action path, while the detailed signal evaluation stays inside the Flow.

5. PROJECT PLANNING & SCHEDULING

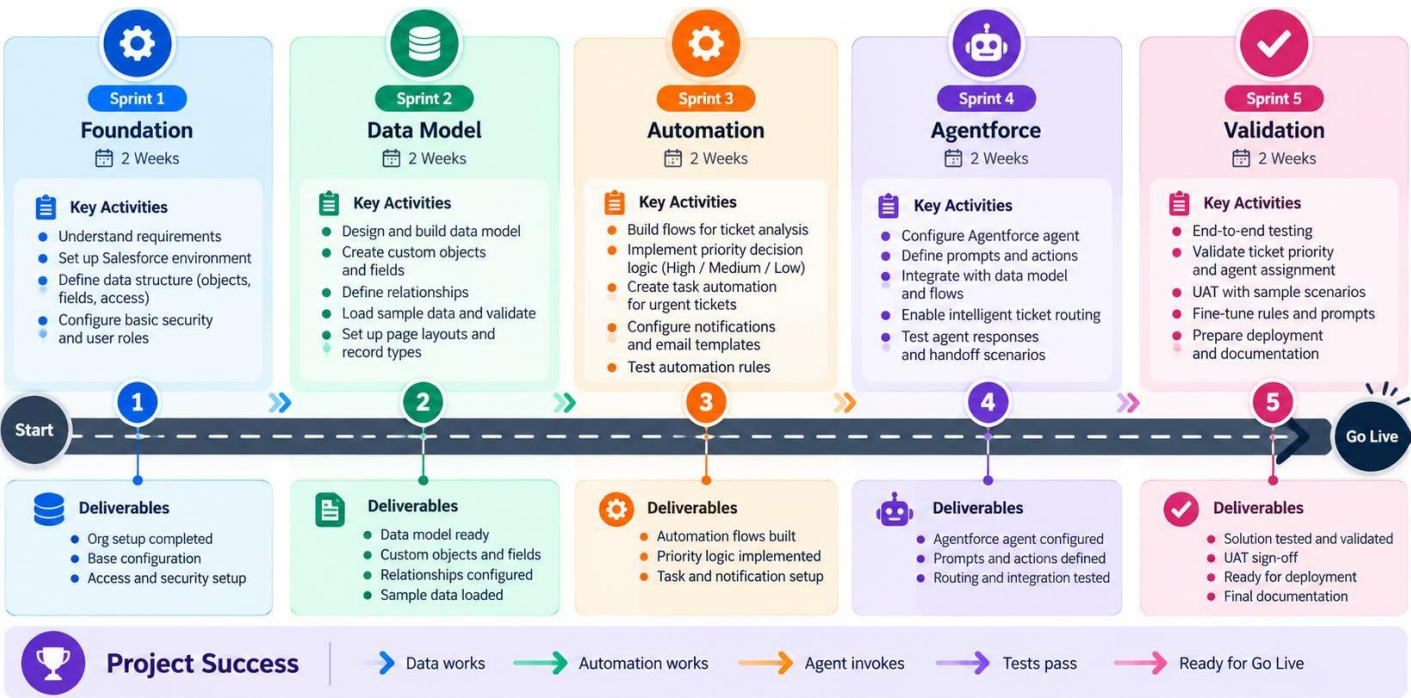
5.1 Agile Planning

- Build the record model before connecting conversational actions.
- Implement and test the decision Flow independently.
- Validate each priority band with controlled test data.
- Connect Agentforce only after the backend action contract is stable.
- Use source control for Salesforce metadata and repeatable deployment.

5.2 Product Backlog

ID	Epic	User Story / Task	Points
US-01	Environment	Configure Salesforce project, permissions and development workflow	3
US-02	Data Model	Create the Support Ticket Intelligence object	5
US-03	Data Model	Create customer and contact relationships	3
US-04	Automation	Build the Auto-Launched Flow skeleton	5
US-05	Decisioning	Implement priority and assignment bands	5
US-06	Agentforce	Create the Support Ticket Priority Analysis subagent	8
US-07	Action	Create High-priority Task and structured outputs	5
US-08	SLA	Add optional age and SLA-risk evaluation	5
US-09	Testing	Validate conversation, Flow outputs and task creation	5
US-10	Version Control	Retrieve metadata and push the working source to GitHub	3

5.3 Sprint Roadmap



5.4 Project Tracking

Sprint	Main Deliverable	Validation Point
Sprint 1	Salesforce environment	Org access, project source and permissions verified
Sprint 2	Custom object and fields	Records can be created and related correctly
Sprint 3	Flow automation	Priority bands and task behavior verified
Sprint 4	Agentforce integration	Conversation successfully invokes the Flow action
Sprint 5	Testing and refinement	High / Medium / Low scenarios and edge cases reviewed

6. PROJECT DEVELOPMENT PHASE

6.1 Salesforce Environment

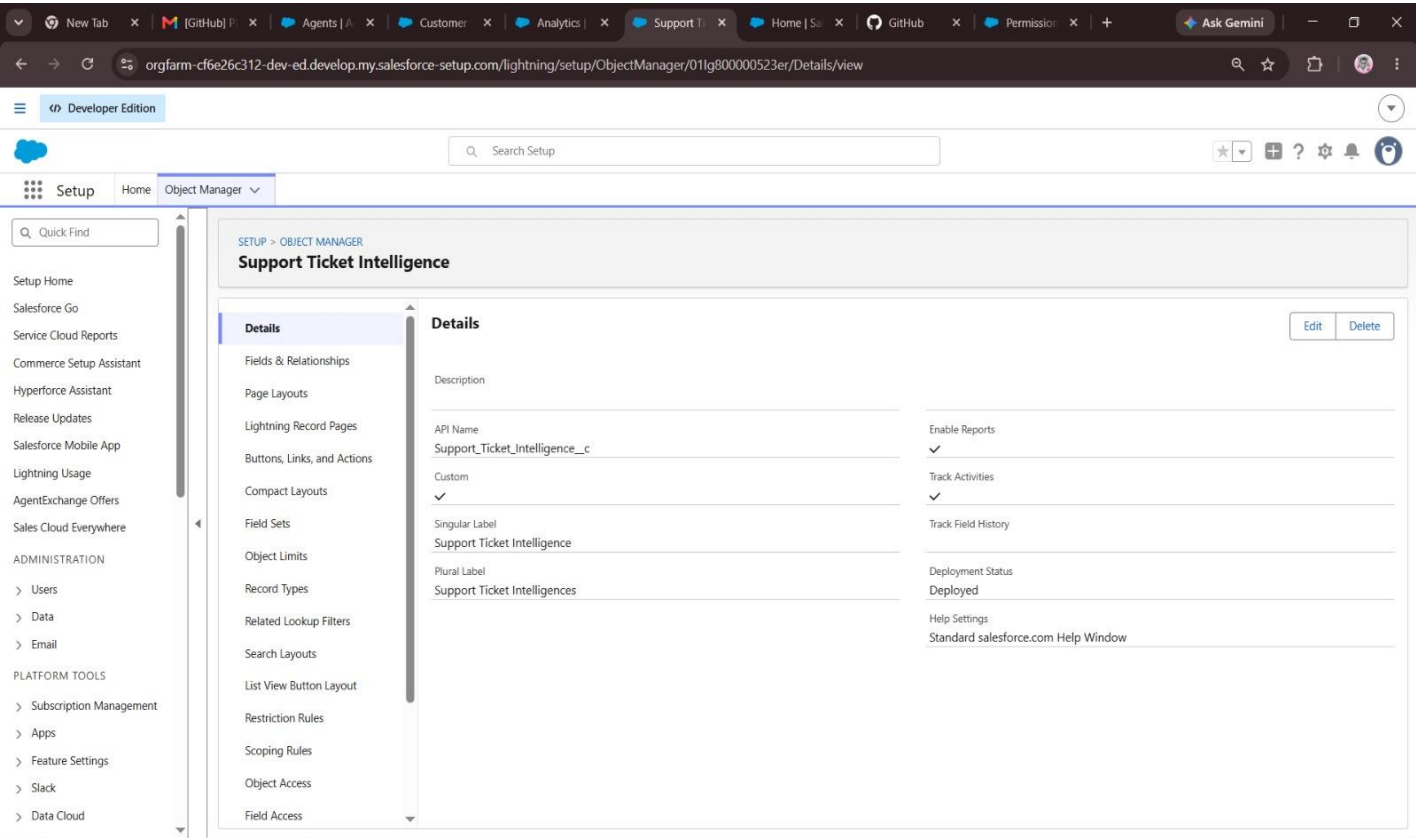
The development workflow uses Salesforce and a Salesforce DX project. Metadata is

retrieved locally, reviewed through the source tree and synchronized with GitHub. This makes changes to objects, flows and Agentforce configuration easier to compare and reproduce across environments.

Component	Project Value
Salesforce Org	Runtime configuration, records and testing
Salesforce DX Project	Local source representation
Salesforce CLI	Retrieve, inspect and deploy metadata
Git	Version control for source changes
GitHub	Remote repository and collaboration point

6.2 Custom Object

Property	Value
Label	Support Ticket Intelligence
API Name	Support_Ticket_Intelligence__c
Purpose	Stores customer support context and the operational values used by the priority workflow.



6.3 Fields and Relationships

Field	API Name	Type	Purpose
Ticket Number	Ticket__Number__c	Auto Number	Unique ticket reference

Customer	Customer__c	Lookup(Account)	Connects the ticket to the customer account
Contact	Contact__c	Lookup(Contact)	Connects the ticket to the requester
Field	API Name	Type	Purpose
Issue Type	Issue_Type__c	Picklist	Classifies the request category
Description	Description__c	Long Text Area	Stores the issue narrative used during evaluation
Priority Level	Priority_Level__c	Picklist	Stores Low, Medium or High
Status	Status__c	Picklist	Tracks New, In Progress or Resolved
Created Date	Created_Date__c	Date	Records the ticket date
Assigned To	Assigned_To__c	Lookup(User)	Stores the support owner
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Flags a possible service-risk condition
Resolution Time	Resolution_Time_hrs__c	Number	Stores resolution duration for later analysis

New Tab x [GitHub] x Agents | x Customer x Analytics | x Support | x Home | x GitHub x Permission x + Ask Gemini

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Developer Edition

Search Setup

Setup Home Object Manager

Quick Find

Setup Home
Salesforce Go
Service Cloud Reports
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Salesforce Mobile App
Lightning Usage
AgentExchange Offers
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
PLATFORM TOOLS
Subscription Management
Apps
Feature Settings
Slack
Data Cloud
Heroku
Mulesoft
Einstein
Objects and Fields
Object Manager
Picklist Value Sets
Schema Builder
Events
Process Automation
User Interface

SETUP > OBJECT MANAGER

Support Ticket Intelligence

Details

Fields & Relationships

15 items. Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Assigned To	Assigned_To__c	Text(255)		
Contact	Contact__c	Lookup(Contact)		✓
Created By	CreatedById	Lookup(User)		
Created Date	Created_Date__c	Date		
Customer	Customer__c	Lookup(Account)		✓
Description	Description__c	Long Text Area(32768)		
Issue Type	Issue_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Priority Level	Priority_Level__c	Picklist		
Resolution Time	Resolution_Time__c	Number(18, 0)		
Resolution Time (hrs)	Resolution_Time_hrs__c	Number(10, 2)		
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox		
Status	Status__c	Picklist		
Ticket Number	Name	Auto Number		✓

6.4 Flow Design

Flow type: Auto-Launched Flow. Input: Account Name. Outputs: Account Id, Ticket Id, Priority Level, Assigned Agent or Queue, and Action Message.

Variable	Type	Purpose
varAccountName	Text / Input	Account identifier supplied by Agentforce
varAccountId	Text / Output	Resolved customer Account Id
varTicketId	Text / Output	Relevant Support Ticket Id
varPriorityLevel	Text / Output	Calculated High / Medium / Low band
varAssignedTo	Text / Output	Support handling level or queue
varActionMessage	Text / Output	Human-readable handling result

6.5 Flow Implementation

Stage	Implementation
1. Account Lookup	Resolve the requested Account and store its identifier.
2. Context Check	Confirm that a usable customer record is available.
3. Ticket Lookup	Retrieve the most relevant recent Support Ticket.
4. Signal Collection	Read description, issue type, age and risk indicators.
5. Priority Decision	Evaluate the signal band and assign High, Medium or Low.
6. High Handling	Create the urgent follow-up Task for the High path.
7. Assignment	Set the configured support level or queue.
8. SLA Check	Evaluate the optional age / SLA risk condition.
9. Final Message	Return the selected action and identifiers.

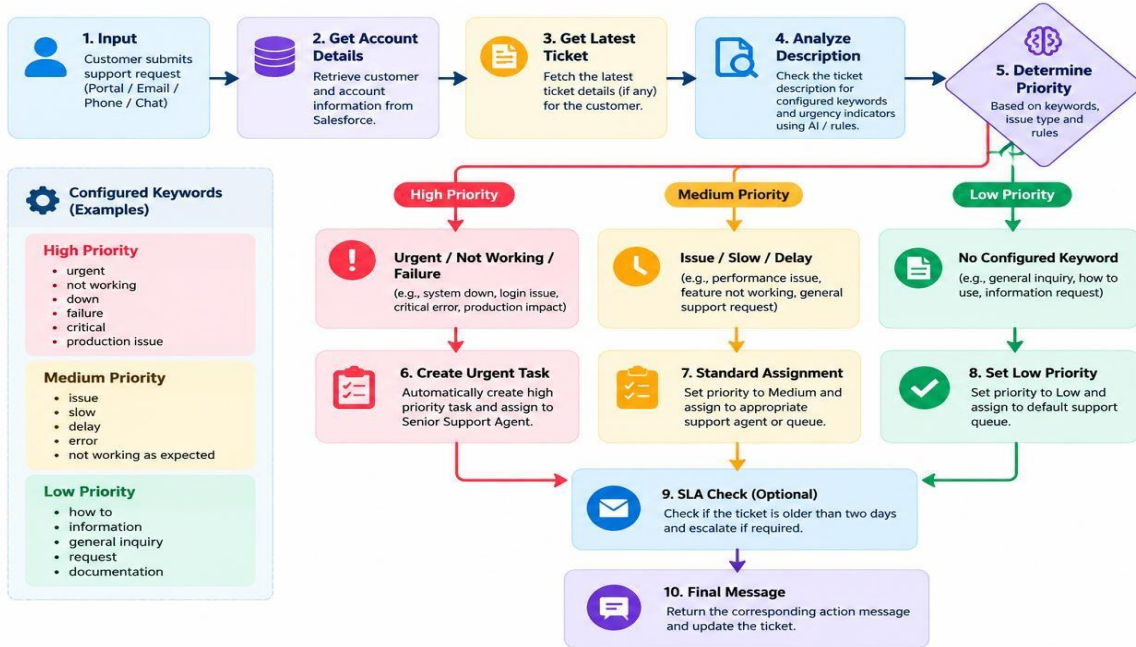
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Priority Automation Logic

Automated ticket analysis and priority assignment based on description and keywords



Smart Analysis.
Right Priority.
Faster Resolution.

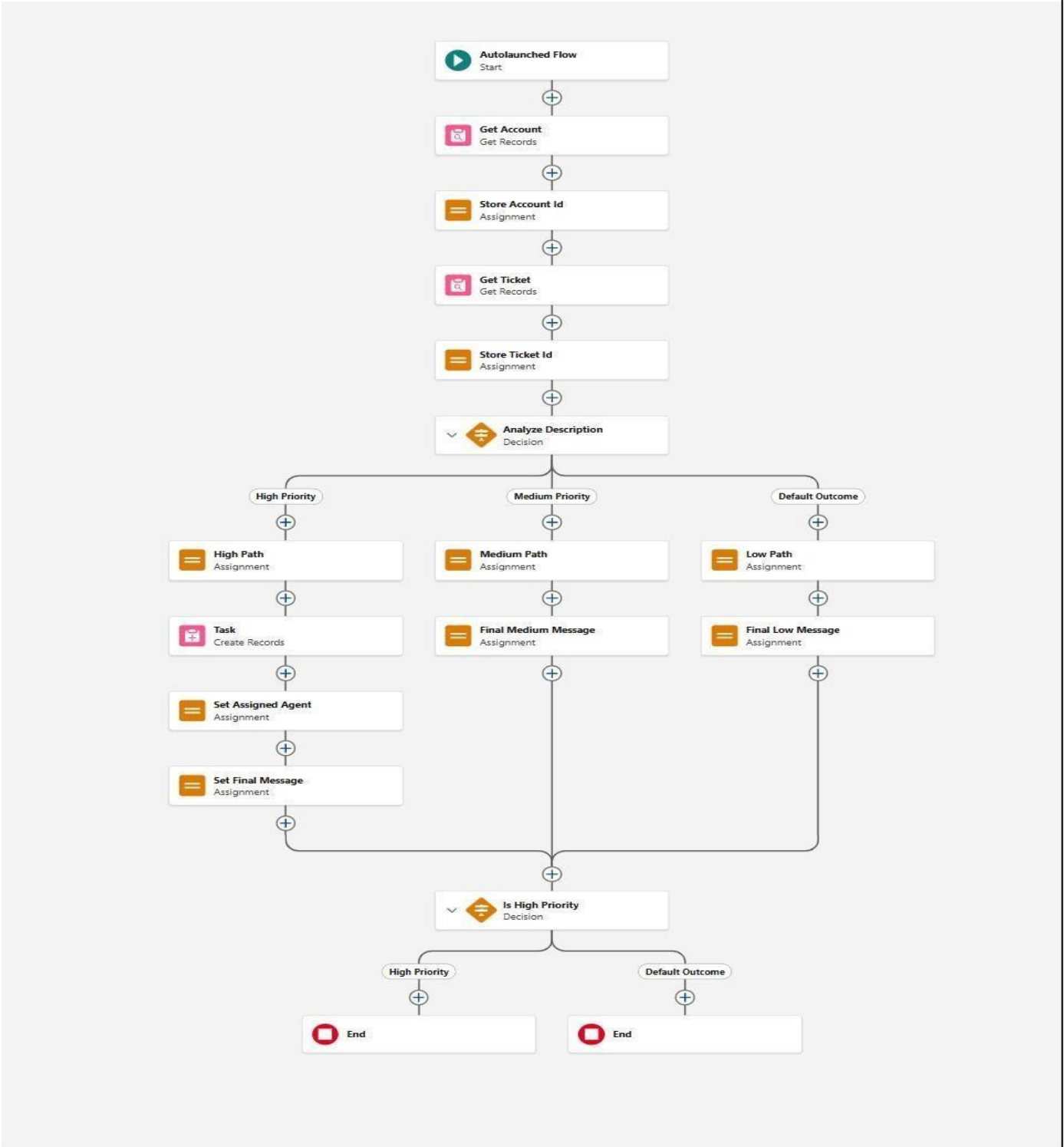


Key Benefits

- Faster response to critical issues
- Accurate and consistent priority assignment
- Reduced manual effort for support agents
- Improved SLA compliance
- Better customer satisfaction

6.5 Flow Implementation — Salesforce Flow View

The implementation screenshot is retained as requested. It shows the configured Flow canvas and its branching structure; the surrounding documentation has been rewritten to describe the revised decision model.



6.6 Agentforce Configuration

The Agentforce configuration uses a bounded subagent named Support Ticket Priority Analysis. Its responsibility is limited to collecting the account context, requesting ticket analysis and presenting the structured result. It should not independently modify unrelated billing, subscription or account settings.

6.7 Agentforce Action

Property	Configuration
Action	Support Ticket Intelligence Flow action
Input	varAccountName
Loading Text	Reviewing ticket context and determining the support priority...
Backend	Support_Ticket_Intelligence Auto-Launched Flow
Purpose	Retrieve relevant ticket data, evaluate priority and return handling outputs.

Support Ticket Intelligence

TKT-0001

Ticket Number

TKT-0001

Customer

Meeravathi

Contact

Issue Type

Technical

Description

The application is not working and this is an urgent issue.

Priority Level

Status

New

Created Date

Assigned To

SLA Breach Risk

Resolution Time

Owner

Meeravathi S

Meeravathi S

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Settings Log Out

DISPLAY DENSITY

Comfy

Compact

OPTIONS

Switch to Salesforce Classic

Add Username

Upcoming &

You have an upcoming task

To change what's shown, try changing your filters.

Show All Activities

To Do List

6.8 Input and Output Contract

Direction	Variable	Meaning
Input	varAccountName	Customer Account Name
Output	varAccountId	Resolved customer identifier
Output	varTicketId	Ticket selected for analysis
Output	varPriorityLevel	Calculated priority band
Output	varAssignedTo	Configured support level or queue
Output	varActionMessage	Final handling instruction

7. FUNCTIONAL & PERFORMANCE TESTING

7.1 Test Strategy

Testing covers the complete automation chain: input collection, customer lookup, ticket retrieval, signal evaluation, priority decision, Task creation, assignment output and Agentforce response. Tests are designed around observable outputs rather than the internal wording of the prompt.

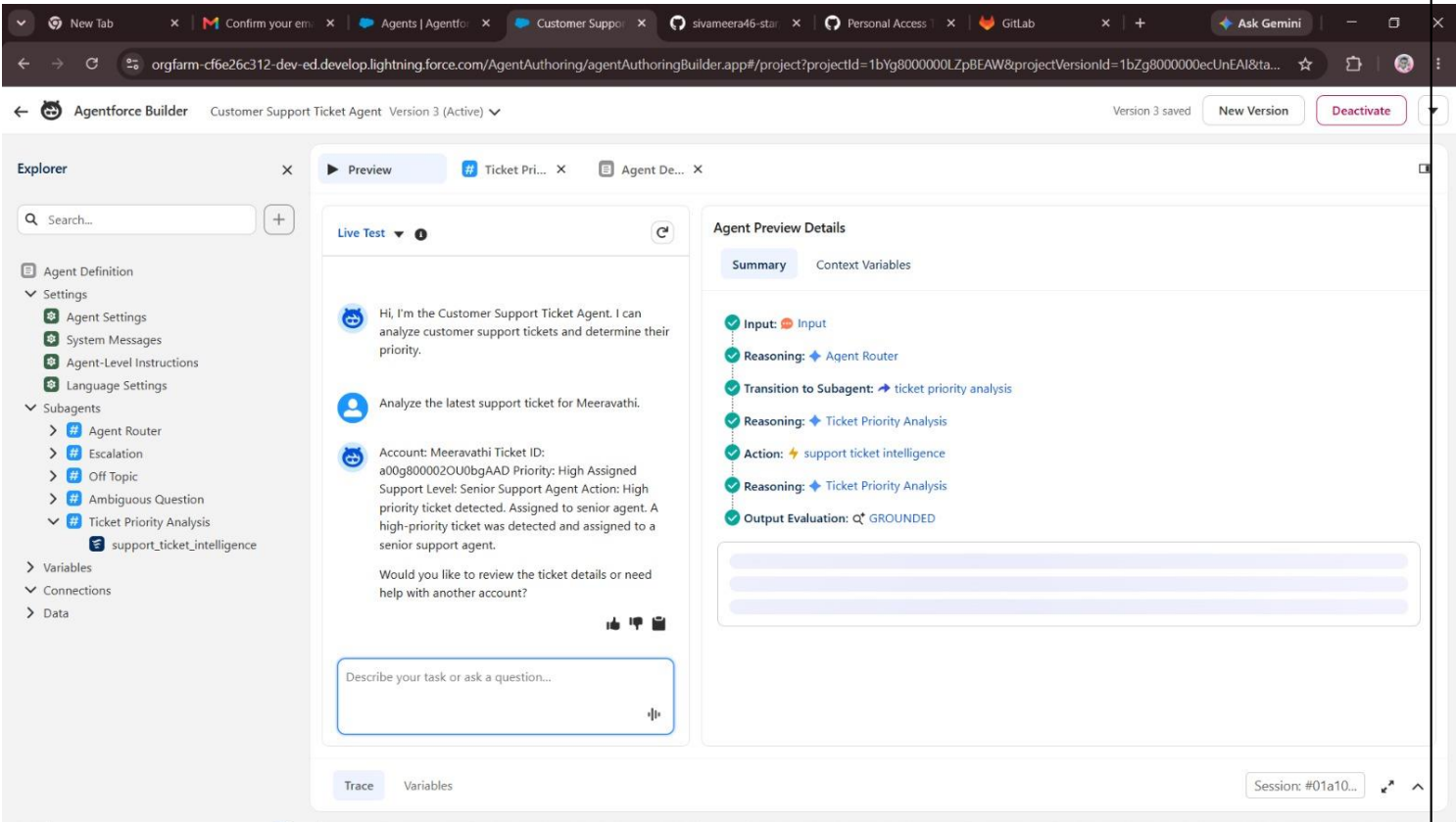
7.2 Test Cases

Test ID	Scenario	Input Condition	Expected Result
TC-01	Critical urgency	Critical signal present	Priority = High
TC-02	Service risk	High SLA-risk condition	Priority = High
TC-03	Operational issue	Issue is meaningful but not critical	Priority = Medium
TC-04	Low urgency	No elevated signal	Priority = Low
TC-05	High task	High priority returned	Urgent Task created
TC-06	Assignment	High / Medium / Low band selected	Configured support path returned
TC-07	Agentforce	Account Name supplied	Structured analysis returned
TC-08	Traceability	Action completes successfully	Ticket and action identifiers available

7.3 High-Priority Task Validation

The High-priority scenario is validated by checking both the decision result and the resulting Salesforce Task. The original validation screenshot is retained; the revised documentation focuses on observable state rather than a single keyword.

Validation	Expected Observation
Task Subject	Urgent Ticket Handling
Priority	High
Status	Not Started
Related Record	Support Ticket Intelligence
Assigned Support Level	Senior Support Agent



7.4 Agentforce Validation

A successful Agentforce test should return the analyzed ticket identifier, calculated priority, assigned support level and action message after the Flow action completes.

7.5 Test Summary

- Customer and ticket retrieval are executed by the backend workflow.
- Priority outcomes follow the configured decision bands.
- High-priority outcomes create traceable urgent work.

8.ADVANTAGES & DISADVANTAGES

7.6 Advantages

- Reduces repetitive first-pass triage.
- Makes the priority decision explicit and reproducible.
- Creates an operational Task for urgent cases.
- Keeps the conversation layer separate from backend record mutation.
- Uses Salesforce records as the authoritative operational context.
- Provides a structured contract that can be tested independently.
- Supports additional assignment and SLA rules without redesigning the interface.
- Can be version-controlled as Salesforce DX metadata.

7.7 Disadvantages

- Rule quality depends on how well urgency and service-risk signals are configured.
- Indirect urgency may require richer context than the current fields provide.
- The relevant-ticket lookup depends on correct customer association.
- Flow and Agentforce configuration require maintenance as business rules change.
- A small set of support levels may not represent complex organizational routing.
- SLA handling remains a decision aid rather than a complete escalation engine.
- Advanced analytics and predictive modeling are outside the current implementation.
- A trained ML model would require historical data, feature design and evaluation.

8.CONCLUSION

The revised design demonstrates an end-to-end Salesforce support workflow in which Agentforce provides conversational access and Salesforce Flow performs controlled operational processing. The workflow resolves customer context, retrieves the relevant ticket, evaluates multiple signals, produces a priority band, selects a support path and creates urgent work when necessary. The architecture also leaves a clear path for richer SLA logic, analytics and future predictive models.

9. FUTURE SCOPE

- Introduce richer urgency, impact and customer-tier signals.
- Add historical-data features for predictive priority scoring.
- Use workload-aware assignment across multiple support queues.
- Add configurable escalation timers and notification policies.

- Expand Agentforce outputs with ticket status, issue type and recommended next actions.
- Add confidence and explanation fields when an ML classifier is introduced.

10. APPENDIX

A. Custom Object

Support_Ticket_Intelligen

ce_c

B. Key Salesforce Records

- Account
- Contact
- Support Ticket Intelligence
- User
- Task

C. Flow

Support_Ticket_Intelligence – Auto-Launched Flow

A. Agentforce

Subagent Support

Ticket Priority

Analysis

D. Agentforce Action

Flow-based Support Ticket Intelligence action with Account Name input and structured ticket-analysis outputs.

E. Revised Visual Set

- Customer support automation ecosystem
- Problem statement and user personas
- Empathy map canvas
- Idea prioritization and customer journey
- Data flow diagram

- Problem–solution fit
- Solution architecture
- Data model design
- Sprint roadmap
- Priority automation logic

Original Salesforce screenshots retained: development field configuration, record/field view, Flow canvas, Agentforce configuration and validation view.

THANK YOU